Reviewer Pack — Polymatroid Rank Deficit in Monotone DNF for k-CLIQUE

Entry 1562169c18c2 · INCONCLUSIVE

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Polymatroid Rank Deficit in Monotone DNF for k-CLIQUE

Entry ID: `1562169c18c2` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-08 22:24:53 UTC

1. Conjecture

Field A (mathematical branch): Polymatroid Theory **Field B** (complexity object): Monotone DNF Formulas

Statement:

> For any k-CLIQUE instance with n variables, the polymatroid rank of its clause hypergraph satisfies rank(H) ≥ Ω(n/k). For any monotone DNF formula with size S(n) = poly(n), the rank(H) is bounded by O(log n).

Rationale (proposer's reasoning):

> Polymatroid rank captures combinatorial structure of clause interactions, which may expose inherent limitations in monotone DNF representations. The conjecture links hypergraph rank to circuit size, leveraging submodularity to isolate k-CLIQUE's complexity.

Taxonomy category: `MONOTONE_CLIQUE` (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `67da669449c15e27`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> (prereg LLM failed:)

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.95	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_clique_cnf(n, k):
        if n < k:
            return None
        variables = list(range(1, n + 1))
        clauses = []
        for i in range(k):
            clause = random.sample(variables, k - i)
            clauses.append(clause)
        return clauses

    def compute_polymatroid_rank(hypergraph):
        rank = 0
        while hypergraph:
            edge = max(hypergraph, key=lambda e: len(e))
            rank += 1
            hypergraph = [e for e in hypergraph if not set(edge).issubset(e)]
        return rank

    def is_monotone_dnf(cnf):
        for clause in cnf:
            if any(var.startswith('-') for var in clause):
                return False
        return True

    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    instances_tested = 0
    conjecture_holds = True
    counterexample = ""

    for n in n_values:
        for _ in range(5):  # Sample 5 instances per n
            cnf = generate_k_clique_cnf(n, random.randint(2, min(4, n // 2)))
            if cnf is None:
                continue
            hypergraph = {tuple(sorted(int(var) for var in clause)) for clause in cnf}
            rank = compute_polymatroid_rank(hypergraph)
            total_rank += rank
            instances_tested += 1
```

```
            # Check polymatroid rank condition
            if rank < n / len(n_values):
                conjecture_holds = False
                counterexample = f"n={n}, rank={rank}"

            # Check monotone DNF condition
            if is_monotone_dnf(cnf) and rank > math.log(n):
                conjecture_holds = False
                counterexample = f"n={n}, rank={rank}"

    mean_rank = total_rank / instances_tested if instances_tested else 0

    return {
        "metric_name": "polymatroid_rank",
        "metric_value": mean_rank,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 87, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 67, in run_trial
    if is_monotone_dnf(cnf) and rank > math.log(n):
```

```
      ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 41, in is_monotone_d
    if any(var.startswith('-') for var in clause):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8e369c5b.py", line 41, in <genexpr>
    if any(var.startswith('-') for var in clause):
           ^^^^^^^^^^^^^^^
AttributeError: 'int' object has no attribute 'startswith'
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Test crashed due to invalid type in clause processing, preventing result evaluation | next: Fix data input format to ensure clauses are string variables, then retest

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	81031
2	preregistration	?	?	0	0	15148
3	novelty	ollama_remote	qwen3:8b	0	0	20743
4	novelty	ollama_remote	qwen3:8b	0	0	13268
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18978
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9326
7	judge	ollama_remote	qwen3:8b	0	0	19134

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 177627 ms total latency. Provider mix: {'ollama_remote': 6, '?': 1}

(full prompt+response transcripts available in `research/audit/1562169c18c2.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/1562169c18c2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/1562169c18c2.tar.gz` (if generated)